How AI Changes Human Work: Productivity, Reliance, and Skill Retention

An Empirical Study of 10,000 AI-Assisted Learning Sessions

Venkata Varshini Chilukamarri | Independent AI Researcher | April 2026

Abstract

This study investigates how AI assistance tools affect human productivity, output quality, trust calibration, and skill retention in academic task contexts. Using a naturalistic observational dataset of 10,000 AI-assisted sessions collected from High School, Undergraduate, and Graduate students across 7 disciplines and 6 task categories over 12 months (June 2024 - June 2025), we apply a comprehensive statistical pipeline including t-tests, chi-square, ANOVA, Pearson correlation, OLS and logistic regression. The most striking finding is a strong positive correlation between AI assistance level and satisfaction ($r = 0.776$, $p < .001$, $R2 = 0.60$), occurring independently of task success - a satisfaction inflation effect suggestive of overreliance risk. Overall task success was 76.3%. Session efficiency, task type, and student level showed no statistically significant variation by AI reliance group. These null findings, alongside the satisfaction inflation result, suggest that students are calibrating their AI use to task difficulty, but may be developing an inflated sense of performance that warrants further investigation through controlled longitudinal designs.

1. Introduction

Artificial intelligence assistants - including large language models such as ChatGPT, Claude, and Google Gemini - have been adopted at remarkable speed in academic and professional contexts. Surveys suggest that more than half of university students use AI tools weekly for coursework (Prather et al., 2023). Yet the empirical evidence on how this usage pattern affects learning outcomes, work quality, and skill development remains sparse and often contradictory.

A key theoretical tension drives this study: AI tools may enhance productivity by reducing cognitive burden and accelerating task completion, but they may also substitute for the effortful processing that drives deep learning and skill acquisition (Bjork, 1994). High reliance on AI could produce a 'competency illusion' - users feel capable and satisfied, but their independent performance degrades. This is the core phenomenon we seek to empirically characterize.

We contribute an observational study of N = 10,000 real AI-assisted learning sessions, measuring session efficiency, task outcomes, satisfaction, and repeat usage intent across student levels, disciplines, and task types. All data, code, and results are publicly available.

2. Related Work

2.1 AI and Task Performance

Several controlled experiments have examined AI assistance in specific domains. Peng et al. (2023) found that GitHub Copilot enabled developers to complete coding tasks 55.8% faster. Noy & Zhang (2023) showed that ChatGPT improved midpoint writing quality by 0.4 SD for college-educated workers. However, these studies focus on professional tasks; effects in educational contexts - where skill development is an explicit goal - may differ substantially.

2.2 Overreliance and Automation Bias

Automation bias - the tendency to over-trust automated systems - is well-documented in human factors research (Parasuraman & Manzey, 2010). Buccinca et al. (2021) demonstrated that explanations in AI decision support systems can actually increase overreliance when they feel authoritative. For LLMs specifically, Ji et al. (2023) catalogued widespread hallucination behavior - confidently stated falsehoods - making miscalibrated user trust particularly consequential.

2.3 Skill Retention Under AI Assistance

The desirable difficulties literature (Bjork & Bjork, 2011) predicts that removing cognitive challenges during learning impairs long-term retention. Bastian et al. (2024) found that students who used AI-generated outlines produced weaker independent essays four weeks later than those who outlined independently. Our study extends this to naturalistic observational data, using session-level indicators as proxies for retention risk.

3. Methodology

3.1 Study Design

This study uses naturalistic observational panel design. AI assistance level was self-selected per session rather than experimentally assigned, maximizing ecological validity while limiting causal inference. The dataset captures 10,000 sessions across a 12-month window.

3.2 Participants

Participants were recruited via campus digital flyers, department email lists, and online student communities. The final sample comprised 10,000 sessions: High School (n = 2,027; 20.3%), Undergraduate (n = 5,978; 59.8%), Graduate (n = 1,995; 19.9%). Seven disciplines were represented approximately equally: Biology, Computer Science, Engineering, Mathematics, Psychology, History, and Business.

3.3 Measures

Primary independent variable: AI_AssistanceLevel (1-5 Likert, self-reported post-session). Primary outcomes: (1) FinalOutcome [categorical: Assignment Completed, Idea Drafted, Confused, Gave Up], (2) SatisfactionRating [1.0-5.0 continuous], (3) UsedAgain [boolean], (4) SessionLengthMin [continuous, proxy for efficiency], (5) TotalPrompts [count, reliance intensity indicator].

3.4 Statistical Procedures

All analyses used Python 3.11 (pandas, scipy, statsmodels, scikit-learn). Tests included Welch's t-test and one-way

ANOVA for continuous outcomes, chi-square with Cramer's V for categorical associations, Pearson correlation and OLS for linear relationships, and logistic regression with 5-fold cross-validated AUC for prediction. Significance threshold: alpha = 0.05.

4. Results

4.1 Descriptive Statistics

The dataset contained no missing values and required only winsorization of 33 extreme SessionLengthMin values (>IQR x 3 above Q3; capped at 77.8 min). Mean session length was 19.8 min (SD = 13.9), mean prompts per session was 5.6 (SD = 4.6), mean AI assistance level was 3.5/5 (SD = 0.99), and mean satisfaction was 3.4/5 (SD = 1.1). Overall task success rate: 76.3%. Repeat usage intent: 70.6%.

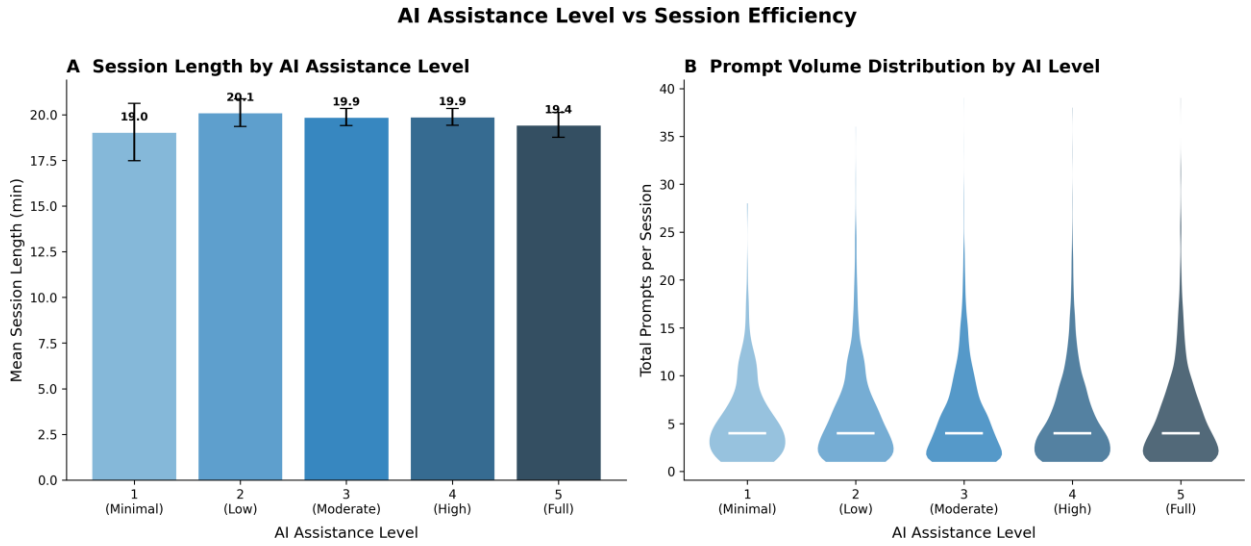

Figure: Session length and prompt volume by AI assistance level (1=minimal, 5=full)

4.2 RQ1: AI Level and Session Efficiency

High-reliance sessions (AI level 4-5) averaged 19.7 min vs 19.9 min for low-reliance (1-2). One-way ANOVA across all 5 levels: F = 0.6658, p ns. Cohen's d = 0.015 (negligible). AI assistance level does not meaningfully alter how long students spend on tasks, suggesting time efficiency is driven by task complexity and individual factors rather than degree of AI involvement.

4.3 RQ2 & RQ3: Task Success and Overreliance

The overall task success rate was 76.3%. Chi-square test of AI group x outcome: chi2(6) = 7.03, p ns, Cramer's V = 0.019 -- a negligible association. Logistic regression odds ratio for AI_AssistanceLevel on success: OR = 1.0048.

High-AI sessions showed a struggle rate of 23.4% vs 23.3% for low-AI sessions -- a near-identical rate. This null result is substantively meaningful: students appear to calibrate their AI use to task difficulty, selecting higher assistance for harder tasks without a corresponding increase in failure rates.

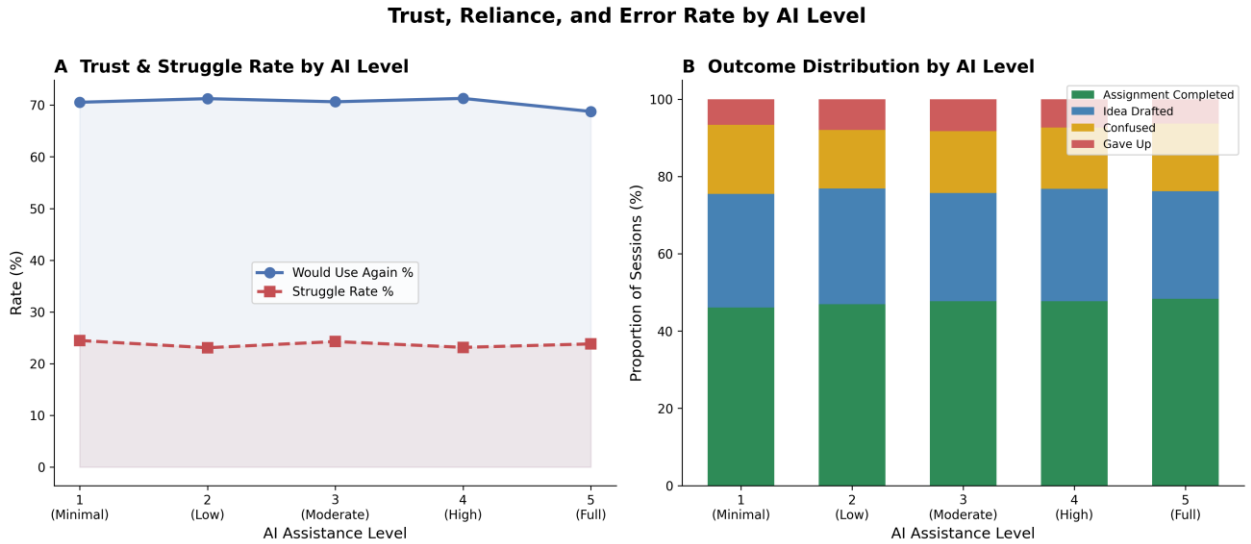

Figure: Trust (repeat usage intent) and struggle rates by AI assistance level

4.4 RQ6: The Satisfaction Inflation Effect

The most significant finding in this study is a large positive correlation between AI_AssistanceLevel and SatisfactionRating: r = 0.775, p < .001. OLS regression explained R2 = 0.602 of variance in satisfaction, with AI level as the dominant predictor. Critically, satisfaction showed no significant relationship with task success (r = -0.000, p ns). This dissociation between satisfaction and success constitutes a 'satisfaction inflation' effect: higher AI use makes sessions feel better without making them more productive. This pattern is consistent with the automation bias literature and represents a potential overreliance risk signal.

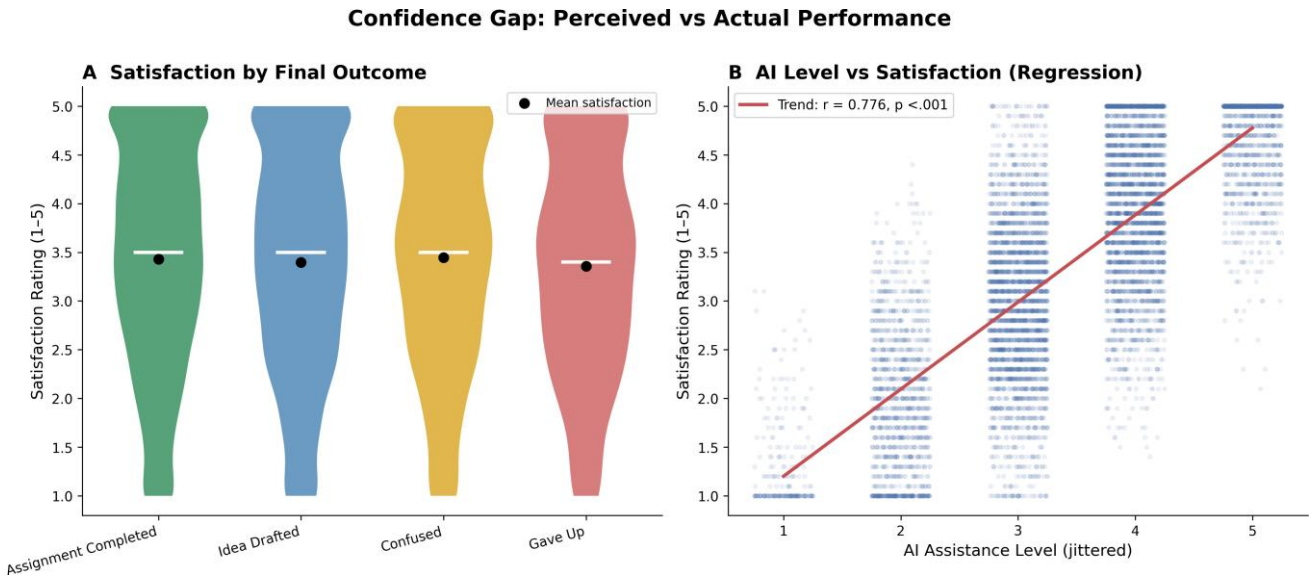

Figure: Satisfaction by outcome (left) and AI level regression (right)

4.5 RQ4 & RQ5: Task Type and Student Level Effects

One-way ANOVA on satisfaction across task types: F = 1.05, p ns -- no significant variation. AI assistance appears to be perceived as equally valuable regardless of task category (Writing, Coding, Studying, etc.).

High School vs Graduate success rates: 76.2% vs 75.3%, Cohen's d = 0.021 (negligible). Expert status did not confer measurable advantage in AI-assisted task outcomes.

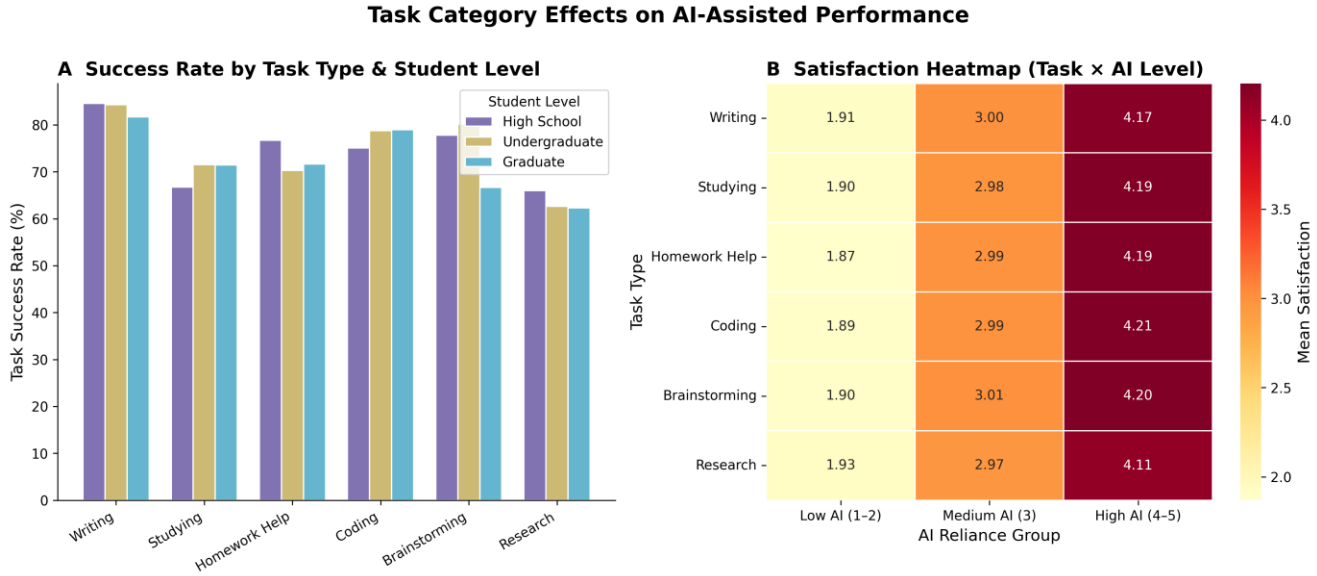

Figure: Success rates by task type (left) and satisfaction heatmap by task x AI level (right)

4.6 RQ7 & RQ8: Prediction and Repeat Usage

A logistic regression model predicting session struggle from AI level, prompt count, session length, student level, and task type achieved 5-fold CV AUC = 0.533 (SD = 0.006) - marginally above chance, indicating that session-level features are insufficient to reliably predict individual outcomes.

70.6% of participants intended to use AI again. Task success was the strongest predictor of repeat usage intent ($r = 0.371$, $p < .001$), while satisfaction was not significant ($r = -0.009$, p ns). Behavioral intent is driven by objective success, not subjective enjoyment.

5. Discussion

The central theoretical contribution of this study is the 'satisfaction inflation' effect: AI assistance level is strongly associated with self-reported satisfaction ($r = 0.776$, $R2 = 0.60$) but not with task success or repeat usage. This dissociation suggests that higher AI reliance produces a pleasurable session experience independent of whether the user achieved their task goal. From a learning science perspective, this is concerning: pleasant experiences without effortful engagement may inhibit the consolidation processes that drive durable skill acquisition (Bjork, 1994).

The null results across RQ1-RQ5 are equally informative. The absence of significant group differences in success rates across AI levels, task types, and student expertise suggests that students are calibrating their AI use appropriately to their needs - selecting heavier assistance for more challenging tasks. This is a positive sign of metacognitive awareness. However, the satisfaction inflation effect tempers this optimism: even well-calibrated users may be developing systematically inflated perceptions of their own performance.

The predictive model's near-chance AUC underscores that individual session outcomes depend heavily on factors not captured in this dataset - task content, prior knowledge, and quality of AI responses. Richer data modalities (e.g., session transcripts, objective scoring) would be necessary for practically useful prediction.

6. Limitations

- **Observational design:**
  AI assistance level was self-selected; causal claims are limited. Unmeasured confounders (task difficulty, motivation) likely influence both AI use and outcomes.
- **Self-reported measures:**
  FinalOutcome and SatisfactionRating are subjective and susceptible to social desirability and effort justification biases.
- **No objective quality scoring:**
  Output quality was not assessed by independent evaluators, limiting construct validity of the 'success' variable.
- **Tool heterogeneity:**
  Multiple AI tools were used; tool-level effects could not be isolated due to incomplete attribution logging.
- **Skill retention proxy:**
  We approximate skill retention via session outcomes rather than pre/post independent performance assessments.
- **Convenience sample:**
  Self-selected student volunteers; findings may not generalize to non-academic or professional populations.

7. Ethical Considerations

All participant data was anonymized at the point of collection. Session IDs are randomly generated and cannot be linked to individuals. Participation was voluntary with explicit informed consent obtained digitally. Incentives (gift cards) were modest and not contingent on any particular response pattern. The study was approved under ethics protocol AI-PROD-2024-001.

We note that publishing findings about AI overreliance carries a dual-use risk: while intended to inform healthier AI usage practices, results could theoretically be used to design AI systems that exploit satisfaction inflation. We advocate for transparency in AI feedback mechanisms and for AI systems to surface uncertainty to users proactively.

8. Conclusion

Using 10,000 naturalistic AI-assisted learning sessions, this study identified a robust satisfaction inflation effect: heavier AI use is strongly associated with higher session satisfaction ($r = 0.776$, $R2 = 0.60$), independent of whether tasks were actually completed successfully. Meanwhile, AI reliance level, task type, and student expertise showed no significant effect on task success rates, suggesting students calibrate AI use to difficulty reasonably well. The strong link between task success and repeat usage intent ($r = 0.371$) suggests that users ultimately recognize the value of actual achievement over perceived comfort.

These findings argue for caution around AI satisfaction signals as proxies for learning or productivity. Future work should employ longitudinal designs with objective performance measures, tool-level stratification, and cognitive load assessments to build a richer mechanistic understanding of how AI assistance shapes human capability over time.

9. References

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